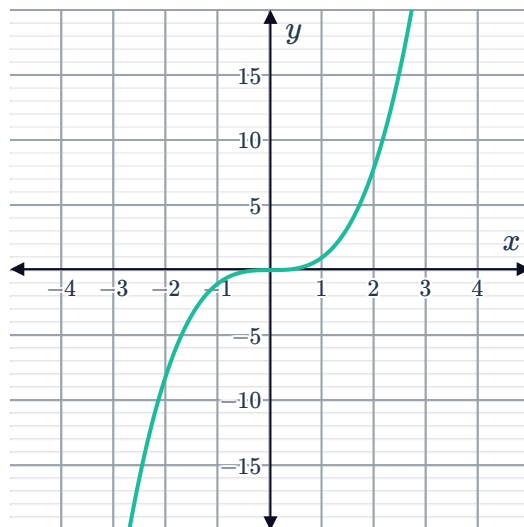


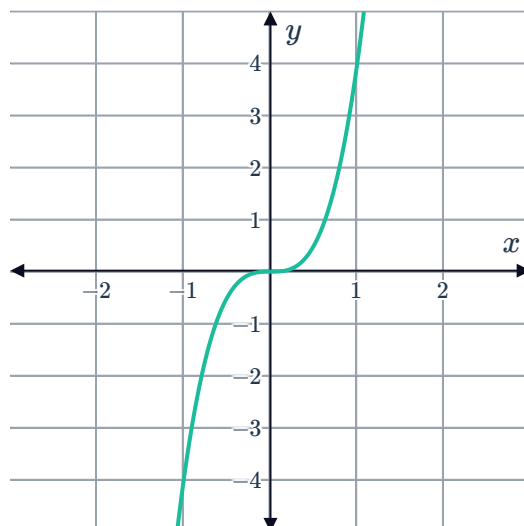
Shape of cubics



- 1 Is the function $y = -2x^3 - 4$ one-to-one?
- 2 By considering the graph of $y = x^3$, determine the following:
 - a As x approaches infinity, what happens to the corresponding y -values?
 - b As x approaches negative infinity, what happens to the corresponding y -values?



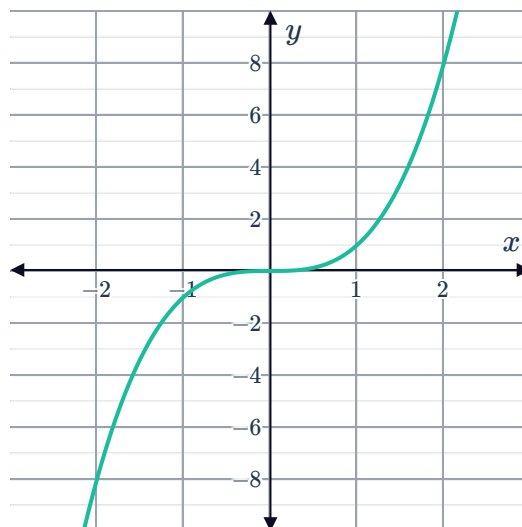
- 3 Determine whether the graphed function shown has an even or odd power. Explain your answer.



4 Consider the graph of the function $y = x^3$

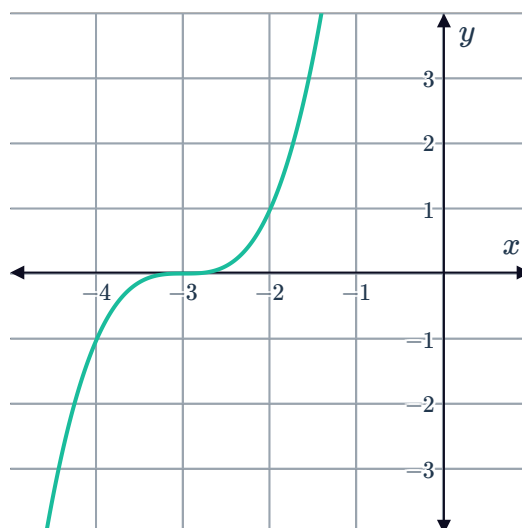


- a For what values of x is the functions concave down?
- b For what values of x is the functions concave up?
- c At what point does the concavity of the curve change?
- d What is this point called?



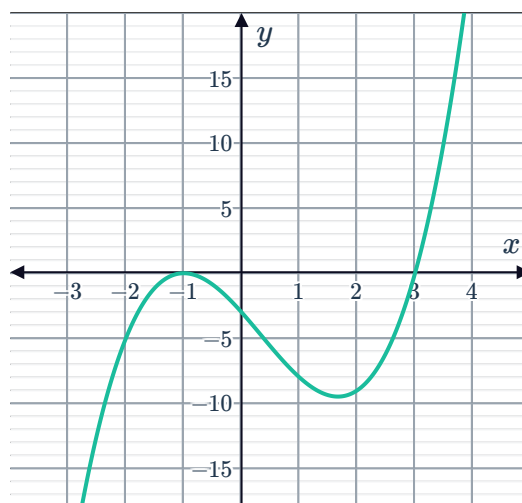
5 Consider the graph of the function:

- a For what values of x is the cubic concave up?
- b For what values of x is the cubic concave down?
- c State the coordinates of the point of inflection.



6 How does the graph of $y = \frac{1}{2}x^3$ differ to the graph of $y = x^3$?

7 Consider the graph of the cubic function shown. For which values of x is $y \geq 0$?





- 8 The graph of $y = x^3$ has a point of inflection at $(0, 0)$. By considering the transformations that have taken place, find the point of inflection of each cubic curve below:

a $y = \frac{2}{3}x^3$

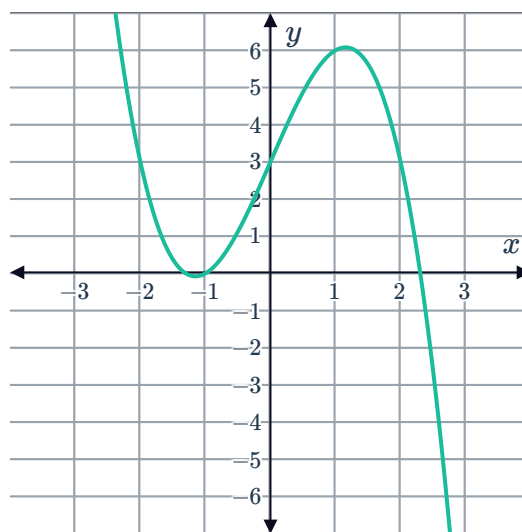
b $y = x^3 + 3$

c $y = -x^3 + 4$

Finding intercepts and equations of cubics

- 9 Consider the graph of the function with equation of the form $y = ax^3 + bx^2 + cx + d$:

- a Is the value of a positive or negative?
- b State the coordinates of the y -intercept.
- c For which values of x is the graph concave up?
- d For which values of x is the graph concave down?
- e State the coordinates of the point of inflection.



- 10 State whether the following functions pass through the origin.

a $y = (x - 2)^2 (x + 3)$

b $y = (x + 1)^3$

c $y = (x - 4)(x + 7)(x - 5)$

d $y = x(x - 6)(x + 8)$

- 11 A cubic function is defined as $y = \frac{1}{2}x^3 + 4$.

- a Find the x -intercept of the function.

- b Find the y -intercept of the function.

- 12 A cubic function has the equation $y = x(x - 4)(x - 3)$. How many x -intercepts does it have?

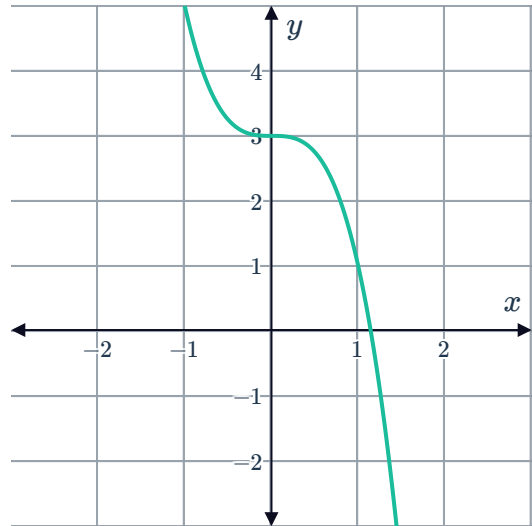
13 Consider the given graph of a cubic function.



- a Determine whether the cubic is positive or negative.
- b State the coordinates of the y -intercept.
- c Which of the following could be the equation of the function?

A $y = -2x^3 + 3$ **B** $y = 2x^3 + 3$

C $y = 2x^3 - 3$ **D** $y = -2x^3 - 3$



14 If the graph of $y = x^3$ is moved to the right by 10 units, what is the new equation?

Graphs of cubics

15 Consider the cubic function $y = -x^3$

- a Complete the following table of values.

x	-2	-1	0	1	2
y					

- b Sketch the graph of the curve.

16 Consider the curve $y = x^3 - 8$.

- a Find the x -intercept.
- b Find the y -intercept.
- c Find the horizontal point of inflection.
- d Sketch the graph of the curve.

17 For each of the functions below:



- i Is the cubic increasing or decreasing from left to right?
- ii Is the cubic more or less steep than the cubic $y = x^3$?
- iii What are the coordinates of the point of inflection?
- iv Sketch the graph.

a $y = 2x^3 + 2$

b $y = -\frac{x^3}{4} + 2$

c $y = -2(x - 2)^3$

d $y = \frac{1}{2}(x - 3)^3$

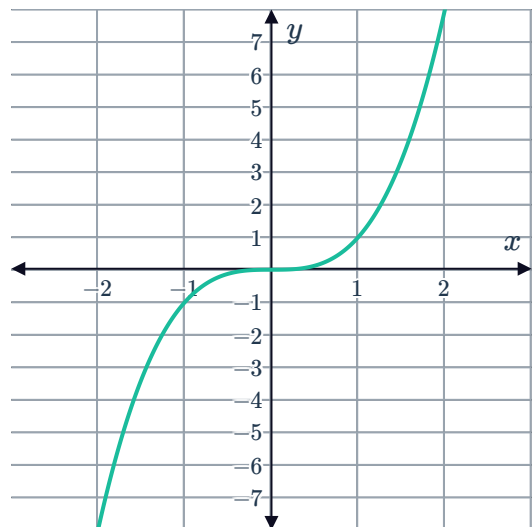
e $y = -\frac{1}{2}(x - 1)^3 + 2$

18 Consider the curve $y = -3(x - 1)^3 + 3$.

- a Find the x -intercept.
- b Find the y -intercept.
- c State the point of inflection.
- d Sketch the graph of the curve.

19 Consider the graph of $y = x^3$.

- a How do we shift the graph of $y = x^3$ to get the graph of $y = (x - 2)^3 - 3$?
- b Hence plot $y = (x - 2)^3 - 3$.



20 Sketch the function $y = (x - 2)(x + 1)(x + 4)$ showing the general shape of the curve and the x -intercepts.

21 For each of the following curves:

- i Find the x -intercept(s).
- ii Find the y -intercept(s).
- iii Sketch the graph of the curve.

a $y = (x + 3)(x + 2)(x - 2)$

b $y = -(x + 4)(x + 2)(x - 1)$

22 Consider the equation $y = 3x + 2x^2 - x^3$.



- a** Express the equation in factorised form.
- b** Find the y -intercept.
- c** Find the x -intercepts.
- d** Sketch the graph of the curve.